WORKED SOLUTION

STEP II 1994, Question 2

Pure • Legendre Polynomials

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s2-q2

The key idea

A factor $(x-a)^m$ survives, slightly weakened, every time we differentiate, so a repeated root of order m is also a root of the first $m-1$ derivatives. The polynomial $(x^2-1)^n = (x-1)^n(x+1)^n$ has repeated roots of order n at both ends of $[-1,1]$, so when we integrate $x^m \mathrm{P}_n(x)$ by parts, each boundary term vanishes. Each integration by parts moves one derivative off $(x^2-1)^n$ and onto x^m, and x^m runs out of derivatives before $(x^2-1)^n$ does.

Solution

Repeated roots and derivatives

By the product rule,

$$\begin{aligned}\mathrm{P}'(x) &= m(x-a)^{m-1}\mathrm{Q}(x) + (x-a)^m\mathrm{Q}'(x)\\ &= (x-a)^{m-1}[m\mathrm{Q}(x) + (x-a)\mathrm{Q}'(x)] = (x-a)^{m-1}\mathrm{R}(x),\end{aligned}$$

where $\mathrm{R}(x) = m\mathrm{Q}(x) + (x-a)\mathrm{Q}'(x)$ is a polynomial, because sums and products of polynomials are polynomials. This is the required form.

The result says that a polynomial with a factor $(x-a)^k$, $k \geq 1$, has a derivative with a factor $(x-a)^{k-1}$. Applying it repeatedly, starting from $k = m$, gives

$$\mathrm{P}^{(r)}(x) = (x-a)^{m-r}\mathrm{R}_r(x) \qquad (0 \leq r \leq m)$$

for some polynomials R_r. Each step is allowed, since we only differentiate $\mathrm{P}^{(r)}$ to get $\mathrm{P}^{(r+1)}$ when $r \leq m-1$, and then the exponent $m-r$ is at least 1. If $1 \leq r \leq m-1$ then $m-r \geq 1$, so $(x-a)$ is still a factor of $\mathrm{P}^{(r)}(x)$, and therefore $\mathrm{P}^{(r)}(a) = 0$. (This also holds for $r = 0$, and in general fails for $r = m$.)

P_n is a polynomial of degree n

Write $F(x) = (x^2-1)^n$. By the binomial theorem, $F(x)$ is x^{2n} plus terms of degree at most $2n-2$. Differentiating x^k n times gives either 0 (if $k < n$) or a multiple of x^{k-n}, so $P_n = F^{(n)}$ is a polynomial of degree at most n. Its x^n term comes only from x^{2n}:

$$\frac{d^n}{dx^n} x^{2n} = 2n(2n-1)\cdots(n+1)\,x^n = \frac{(2n)!}{n!}\,x^n.$$

This coefficient is non-zero, so P_n has degree exactly n.

The vanishing integrals

Since $F(x) = (x-1)^n(x+1)^n$, the first part with $a = 1$, $m = n$ and $Q(x) = (x+1)^n$ shows that $F^{(r)}(1) = 0$ for $1 \le r \le n-1$. The same argument with $a = -1$ gives $F^{(r)}(-1) = 0$. Also $F(\pm 1) = 0$ directly. So

$$F^{(r)}(\pm 1) = 0 \qquad \text{for } 0 \le r \le n-1. \tag{$*$}$$

Now take $j \ge 1$ and $1 \le k \le n$. Integrating by parts, with x^j differentiated and $F^{(k)}$ integrated, gives

$$\begin{aligned}\int_{-1}^{1} x^j F^{(k)}(x)\,dx &= \Big[x^j F^{(k-1)}(x)\Big]_{-1}^{1} - j\int_{-1}^{1} x^{j-1}F^{(k-1)}(x)\,dx \\ &= -j\int_{-1}^{1} x^{j-1}F^{(k-1)}(x)\,dx,\end{aligned}$$

because $k-1 \le n-1$, so the boundary term vanishes by $(*)$.

Ask yourself How many times can we use this before something goes wrong?

Each use lowers the power of x by one and the order of the derivative by one. We start at x^m and $F^{(n)}$, so we may apply it m times. The step needs $k \ge 1$, and the smallest k we use is $n-m+1 \ge 1$. After m steps,

$$\int_{-1}^{1} x^m P_n(x)\,dx = (-1)^m m! \int_{-1}^{1} F^{(n-m)}(x)\,dx.$$

(For $m = 0$ this is just the original integral.)

If $0 \le m \le n-1$ then $n-m \ge 1$, so the last integrand is itself a derivative and

$$\int_{-1}^{1} x^m P_n(x)\,dx = (-1)^m m!\Big[F^{(n-m-1)}(x)\Big]_{-1}^{1} = 0,$$

by $(*)$, since $0 \le n-m-1 \le n-1$. This is the required result.

The case $m = n$

The same reduction with $m = n$ is still valid (the last step uses $k = 1$) and gives

$$\int_{-1}^{1} x^n P_n(x)\,dx = (-1)^n n! \int_{-1}^{1} (x^2-1)^n\,dx = n!\int_{-1}^{1} (1-x^2)^n\,dx,$$

since $(x^2-1)^n = (-1)^n(1-x^2)^n$. The integrand is even, and the substitution $x = \sin t$, $\mathrm{d}x = \cos t\,\mathrm{d}t$, takes $[0,1]$ to $[0, \frac{\pi}{2}]$ with $1 - x^2 = \cos^2 t$. So, using the formula in the hint,

$$\int_{-1}^{1} (1-x^2)^n \,\mathrm{d}x = 2\int_0^{\pi/2} \cos^{2n+1} t \,\mathrm{d}t = \frac{2^{2n+1}(n!)^2}{(2n+1)!}.$$

Therefore

$$\boxed{\int_{-1}^{1} x^n \mathrm{P}_n(x) \,\mathrm{d}x = \frac{2^{2n+1}(n!)^3}{(2n+1)!}.}$$

Checks

For $n = 1$, $\mathrm{P}_1(x) = 2x$, so $\int_{-1}^{1} x \cdot 2x \,\mathrm{d}x = \frac{4}{3}$, and the formula gives $\frac{8\cdot 1}{6} = \frac{4}{3}$. For $n = 2$, $\mathrm{P}_2(x) = 12x^2 - 4$. Then $\int_{-1}^{1}(12x^2-4)\,\mathrm{d}x = 8 - 8 = 0$ and $\int_{-1}^{1} x(12x^2-4)\,\mathrm{d}x = 0$ (an odd integrand), as predicted. Also $\int_{-1}^{1} x^2(12x^2-4)\,\mathrm{d}x = \frac{24}{5} - \frac{8}{3} = \frac{32}{15}$, which matches $\frac{2^5 \cdot 8}{120} = \frac{32}{15}$. We checked all of this exactly by computer for $n \le 10$.

Common slips

- Forgetting that the boundary terms need $\mathrm{F}^{(r)}(\pm 1) = 0$ for $r = 0$ as well as $1 \le r \le n-1$. The final step for $m = n-1$ uses $r = 0$.
- Losing the sign: $(x^2-1)^n = (-1)^n(1-x^2)^n$ cancels the $(-1)^n$ from the integrations by parts, so the answer is positive. That is reasonable, since $x^n \mathrm{P}_n(x)$ has positive leading coefficient.

Beyond STEP

Up to a constant, P_n is the Legendre polynomial of degree n, and this formula for it is Rodrigues' formula. The first result says P_n is orthogonal to every polynomial of lower degree with respect to the inner product $\int_{-1}^{1} fg\,\mathrm{d}x$. This is the property behind Legendre series and Gaussian quadrature. Legendre polynomials also turn up in the solution of Laplace's equation in spherical coordinates. Boyce and DiPrima, *Elementary Differential Equations*, meets them through Legendre's differential equation.

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